Nathan Riek

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SUMMARY

PhD-trained machine learning engineer with 5+ years building scalable data pipelines, machine learning models, and user-friendly applications, with deep experience in physiological timeseries data.

EXPERIENCE

Founding Machine Learning Engineer – IntelliCardio
Sep 2025 – Present

- Built a desktop application that plots ECGs, deidentifies ECG files, and generates median beats, features, and machine learning model predictions for non-technical users

Postdoctoral Research Associate – University of Rochester
Sep 2025 – Present

- Deploying a machine learning–driven decision support tool into a production workflow within an Azure cloud environment
- Modified ECG preprocessing pipeline from PhD research for compatibility with the AliveCor KardiaMobile 6L at-home device

Data Science Intern – HeartBeam
Sep 2024 – Dec 2024

- Trained deep learning models in PyTorch to estimate cardiac age from noisy ECG time-series data
- Automated data preparation and prediction retrieval using Selenium, minimizing manual effort and facilitating scalable analysis of ECG datasets

PhD Researcher – University of Pittsburgh
Jan 2021 – Aug 2025

- Designed scalable ECG and EEG preprocessing pipelines using Python and MATLAB, adopted by 5+ research institutions
- Developed PyTorch models for ECG-based classification of occlusion myocardial infarction, winning the 2024 ISCE Early Career Investigator Award
- Built an explainable machine learning decision support platform in Dash and Azure to assist clinicians with ECG interpretation
- Co-authored publications in top-tier journals including Nature Medicine, European Heart Journal, and IEEE Transactions on Biomedical Engineering
    - Google Scholar: https://scholar.google.com/citations?user=92lDKDSlhugC&hl
    - ORCID: https://orcid.org/0000-0003-3286-560X

Engineering Co-Op - Eaton
Sep 2018 – Aug 2020

- Built a C# GUI test automation tool, integrated RF instrumentation into TestStand/LabVIEW workflow, and contributed to Lean Six Sigma initiatives

EDUCATION

PhD, Electrical and Computer Engineering – University of Pittsburgh, 2025
BS, Electrical Engineering – University of Pittsburgh, 2020
IBM Data Science Professional Certificate – Coursera, 2025
IBM Generative AI Engineering Professional Certificate – Coursera, 2025

SKILLS

Machine Learning & Modeling: PyTorch, Scikit-learn
Programming: Python, R, MATLAB
Data Engineering & Automation: Pandas, NumPy, SciPy, SQL, Selenium, LangChain
Application Development: Dash, Streamlit, Gradio, PySide6
Visualization: Plotly, Matplotlib, Seaborn
Cloud & Deployment: Azure, AWS, Docker
Version Control: Git, GitHub